

Bond Duration and Convexity Revisited

Second-Order Price Approximations

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Abstract

We revisit approximations of bond price changes built from modified duration D and convexity C . The previously proposed self-consistent closed form, obtained by replacing P_0 with the updated price P inside the Taylor expansion, is shown to be accurate only to first order and, in numerical tests, less accurate than the standard duration-plus-convexity approximation. We give two alternatives that are correct to second order: a rational (Padé) form and a log-price form. On a panel of 465 bond-shock pairs the log form reduces RMSE by a factor of about 18 relative to duration-plus-convexity.

The paper ends with “The End”

1 Setup

For a bond with cash flows C_t at times $t = 1, \dots, T$ and flat annual yield y , the price is

$$P(y) = \sum_{t=1}^T \frac{C_t}{(1+y)^t}. \quad (1)$$

Modified duration and convexity are the normalised first and second derivatives at the base yield y_0 [1, 2, 3]:

$$D = -\frac{1}{P_0} \frac{dP}{dy} \Big|_{y_0} = \frac{1}{P_0(1+y_0)} \sum_t \frac{t C_t}{(1+y_0)^t}, \quad (2)$$

$$C = \frac{1}{P_0} \frac{d^2 P}{dy^2} \Big|_{y_0} = \frac{1}{P_0} \sum_t \frac{t(t+1) C_t}{(1+y_0)^{t+2}}. \quad (3)$$

Since every term of P is a positive convex function of y , P is strictly decreasing and strictly convex, and $C > 0$. The standard second-order approximation, with $\Delta y = y - y_0$, is

$$P \approx P_0 \left(1 - D \Delta y + \frac{1}{2} C \Delta y^2 \right). \quad (4)$$

Because P is convex, the tangent line lies below the curve [4], so the duration-only error (approximation minus exact) is negative for every nonzero shock, of either sign.

2 The self-consistent form is first-order only

The original paper substitutes $P_0 = P - \Delta P$ into the right side of (4) and solves for ΔP , which yields

$$P \approx \frac{P_0}{1 + D \Delta y - \frac{1}{2} C \Delta y^2}. \quad (5)$$

This step is not an identity: D and C are derivatives at y_0 , and the Taylor coefficients multiply P_0 , not the unknown P . Expanding (5) gives

$$\frac{P_0}{1 + D\Delta y - \frac{1}{2}C\Delta y^2} = P_0 \left[1 - D\Delta y + \left(\frac{1}{2}C + D^2\right)\Delta y^2 + O(\Delta y^3) \right], \quad (6)$$

whose quadratic coefficient is wrong by $D^2\Delta y^2$, the same order as the convexity correction itself. It therefore matches the true expansion only to first order, which is the accuracy of duration alone.

3 Two correct second-order alternatives

Proposition 1 (Padé form). *Let*

$$P_R = \frac{P_0}{1 + D\Delta y + \left(D^2 - \frac{1}{2}C\right)\Delta y^2}. \quad (7)$$

Then $P_R - P(y_0 + \Delta y) = O(\Delta y^3)$.

Proof. For $1/(1 + a\Delta y + b\Delta y^2) = 1 - a\Delta y + (a^2 - b)\Delta y^2 + O(\Delta y^3)$, matching coefficients with $1 - D\Delta y + \frac{1}{2}C\Delta y^2$ gives $a = D$ and $b = D^2 - \frac{1}{2}C$. This is the $[0/2]$ Padé approximant of P/P_0 [5]. $\square$

Proposition 2 (Log-price form). *Since $\frac{d^2}{dy^2} \ln P = \frac{P''}{P} - \left(\frac{P'}{P}\right)^2$, at y_0 we have $\frac{d}{dy} \ln P = -D$ and $\frac{d^2}{dy^2} \ln P = C - D^2$. Hence*

$$P_{\log} = P_0 \exp\left(-D\Delta y + \frac{1}{2}(C - D^2)\Delta y^2\right) \quad (8)$$

is also second-order accurate.

Remark 1. *The denominator of (7) can vanish for large shocks, so (8) is the safer default. Both reduce to (4) to second order, and both are positive-valued in the log form, unlike the quadratic (4).*

4 Numerical results

Take a 10-year, 6% annual-coupon par bond with $F = 1000$, $y_0 = 6\%$. Direct computation gives $P_0 = 1000$, $D = 7.360$, and $C = 69.74$.

Δy (bps)	Exact	Dur. only	Dur.+Conv.	Padé	Log	Orig. (5)
-300	1255.91	-35.10	-3.72	-0.52	-0.06	+81.32
-100	1077.22	-3.62	-0.13	-0.01	-0.00	+6.31
-50	1037.69	-0.89	-0.02	-0.00	-0.00	+1.46
+50	964.06	-0.86	+0.02	+0.00	+0.00	+1.26
+100	929.76	-3.37	+0.12	+0.01	+0.00	+4.72
+300	807.47	-28.27	+3.11	+0.17	+0.04	+33.28

Table 1: Exact price and approximation error (approximation minus exact, in dollars). Duration-only errors are negative on both sides, as convexity requires.

For a wider test we use the grid of coupons $\{2, 5, 8\}\%$, base yields $\{3, 5, 7, 10\}\%$, maturities $\{2, 5, 10, 20, 30\}$ years, and shocks $\{\pm 50, \pm 100, \pm 200, \pm 300\}$ bps, dropping cases with $y_0 + \Delta y \leq 0$, which leaves 465 pairs.

Method	RMSE (\$)	MAE (\$)	Max abs. error (\$)
Duration only	40.87	18.86	263.7
Duration + convexity (4)	9.22	3.26	73.9
Original (5)	120.26	38.18	1427.9
Padé (7)	1.41	0.40	15.9
Log form (8)	0.52	0.15	4.8

Table 2: Error metrics over the 465-pair panel.

5 Discussion

The log form is the most accurate because $\ln P$ is much closer to quadratic in y than P itself, and it costs one exponential. All three corrected formulas still use only P_0 , D , and C . Remaining error is dominated by the omitted third derivative, so for very long maturities and shocks beyond ± 300 bps full repricing is preferable. Duration and convexity as regression slopes and portfolio aggregation follow standard treatments [2, 3]. The same expansions underlie hedging and immunisation arguments [1]. Embedded options can make C negative, in which case (4) and (8) still apply but (7) should be checked for sign changes of the denominator.

References

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